🏥 Sehat-Assist AI

Pakistan's First Autonomous Multi-Agent Healthcare Assistant

Built on Qoder — A Complete End-to-End Medical Navigation & Safety Platform

🎯 Problem Statement

Every day in Pakistan, millions of people face critical healthcare roadblocks that cost time, money, and sometimes lives:

- Patients unknowingly consume fake or substandard medicines (Jali adviyat).
- Families receive complex lab reports in English medical jargon and panic, not understanding what it means.
- Patients waste hours standing in the wrong hospital department queues.
- People struggle to find the nearest specialized hospital in emergency situations.
- Appointments are missed, and patients forget to take life-saving medications on time.

> **Sehat-Assist AI solves all of this through an autonomous, voice-first, multi-agent AI system.**

💡 What We Built

Sehat-Assist AI is an advanced **multi-agent AI ecosystem** designed for the common Pakistani. It takes unstructured inputs (voice symptoms, images of medicines, lab reports, or prescriptions) and autonomously:

1. **Verifies** the authenticity of medicine boxes to prevent counterfeit drug usage.
2. **Translates & Simplifies** medical lab reports into local language voice notes (Urdu, Sindhi, Punjabi, Pashto) without making medical diagnoses.
3. **Triages** symptoms to accurately recommend the correct medical department.
4. **Locates** the nearest appropriate hospital using real-time GPS and mapping APIs.
5. **Books** appointments by physically making an AI voice call to the hospital via Twilio API.
6. **Generates** a digital E-Parchi (appointment slip) for the patient.
7. **Schedules** automated reminders for medicine intake based on prescription scans.

🤖 Agent Architecture

```
USER INPUT
(Voice Symptoms / Camera Scans: Meds, Reports, Prescriptions)
                    ↓
           ┌─────────────────────┐
           │  MAIN ORCHESTRATOR  │
           │      AGENT        │
           │  Intent Detection  │
           │  Permission Mgmt   │
           └─────────────────────┘
              ↓    ↓    ↓    ↓
     ┌─────────┐ ┌──────┐ ┌────────┐ ┌───────────┐
     │ PHARMA  │ │LINGO │ │ TRIAGE │ │ CARE-SYNC │
     │ CHECK   │ │-MED  │ │ AGENT  │ │   AGENT   │
     │ AGENT   │ │AGENT │ │        │ │           │
     └─────────┘ └──────┘ └────────┘ └───────────┘
                            ↓
                      ┌────────┐
                      │  GEO   │
                      │LOCATOR │
                      └────────┘
                            ↓
                      ┌───────────┐
                      │AUTO-BOOK &│
                      │ E-PARCHI  │
                      └───────────┘
                            ↓
              FINAL RESPONSE & ACTION TO USER
```

🧠 Agent Descriptions

1. Main Orchestrator Agent

Role: The Traffic Controller

Responsibilities:

- Ask for device permissions (Microphone, Camera, Location) on first launch.
- Analyze the user's input (Is it a picture of a pill box? A lab report? A voice note?).
- Route the request to the exact specialized agent needed.

- Maintain session context.

2. Pharma-Check AI (Fake Medicine Detector)

Role: The Quality Inspector

Responsibilities:

- Activate device camera to scan medicine packaging.
- Extract barcodes, QR codes, DRAP registration numbers using Computer Vision.
- Compare anomalies against a database.
- Issue clear warnings.


```
{ "scanned_item": "Panadol CF", "drap_number": "012345", "authenticity_status":
"WARNING", "reasoning": "Batch number format mismatch. DO NOT CONSUME." }
```


3. Lingo-Med AI (Report Simplifier)

Role: The Friendly Guide

Responsibilities:

- Process uploaded lab reports using OCR.
- Identify out-of-range parameters.
- Generate a comforting, highly simplified explanation via TTS in regional languages.
- **Strict Guardrail:** Append the "Assist, not Diagnose" disclaimer.

4. Smart Triage AI

Role: The Reception Desk

Responsibilities:

- Listen to patient symptoms in natural language.
- Process keywords to determine exact medical specialty (e.g., Cardiology, Gastroenterology).

- Flag emergency keywords for immediate red alerts.

5. Geo-Med Locator AI

Role: The Navigator

Responsibilities:

- Fetch real-time GPS coordinates.
- Query Google Maps API to find the nearest hospitals equipped with the identified department.
- Return exact distance, estimated travel time, and navigation link.

6. Auto-Booking & E-Parchi Agent

Role: The Executive Assistant

Responsibilities:

- Trigger Twilio's Programmable Voice API.
- Conduct a real-time AI voice conversation to request an appointment slot.
- Adapt to receptionist responses.
- Generate a downloadable "E-Parchi".

SEHAT-ASSIST AI - E-PARCHI

Patient Name: Ali Raza
Date: 21 August 2026
Time: 6:00 PM

Hospital: XYZ Clinic
Department: Gastroenterology
Status: CONFIRMED (Voice)

📍 *Distance: 2.5 km from you*
⚕️ *Show this pass at counter*

7. Care-Sync AI (Prescription Reminder)

Role: The Caretaker

Responsibilities:

- Read doctor prescriptions (parchi).
- Extract medicine name, dosage, and timings.
- Schedule automated push notifications or reminders.

🔄 Complete Flow Example (Triage to Booking)

```
[18:01:00] ORCHESTRATOR: Intent = SYMPTOM_TO_HOSPITAL [18:01:01] TRIAGE AGENT: Symptoms analyzed. Department = Gastroenterology. Urgency = HIGH. [18:01:02] GEO-LOCATOR: GPS = Gulshan-e-Iqbal. Querying Maps API... [18:01:03] GEO-LOCATOR: Nearest facility = Patel Hospital (1.2 km). [18:01:03] ORCHESTRATOR: "Aap ke nazdeek Patel Hospital hai. Appointment milaun?" [18:01:08] USER: "Haan bhai mila do." [18:01:09] AUTO-BOOKING AGENT: Initiating Twilio Voice Call to Prototype number. [18:01:10] TWILIO LOG: AI: "Mujhe pait ke dard ki appointment chahiye." [18:01:15] TWILIO LOG: Receptionist: "6 baje aajaen." [18:01:16] TWILIO LOG: AI: "Shukriya, confirm." Call disconnected. [18:01:17] AUTO-BOOKING AGENT: E-Parchi generated for 6:00 PM. [18:01:18] ORCHESTRATOR: UI updated with E-Parchi.
```

👨‍💻 Team Task Division Plan

To accelerate development, the project is divided into two parallel tracks based on technical domains:

Track A: Vision & Analytics Engine	Track B: Action & Navigation Engine
Focus: Camera, OCR, Image Processing, Cron Jobs	**Focus:** Voice, Telephony, GPS, Mapping
• Pharma-Check AI (Medicine Scanning) • Lingo-Med AI (Report OCR & Simplify) • Care-Sync AI (Prescription Parsing)	• Smart Triage AI (Voice to Department) • Geo-Med Locator (GPS & Maps API) • Auto-Booking Agent (Twilio Voice API)
APIs: Vision API, TTS, Schedulers	**APIs:** Speech-to-Text, Twilio, Google Maps

⚠️ **Important:** Sehat-Assist AI operates strictly on an "Assist, Not Diagnose" policy. All AI-generated simplifications must explicitly state they are for informational purposes only. Simulated bookings in the prototype phase must route to designated testing numbers.